# Integrating Skin Tone Quantification with Pulse Oximetry in a Wearable Device

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***Abstract*—Current pulse oximeters used in clinical settings have error rates up to 12% in hypoxic conditions because skin-tone variability is not adequately accounted for during device development and validation. The algorithms for these devices are optimized on the test population and may be biased toward the skin-tone characteristics of that population, limiting generalizability across broader patient groups. This paper introduces a wearable device that combines multi-spectral spectroscopy with pulse-oximetry sensing, enabling paired measurements for studying skin-tone-dependent error. The sensing suite fits into a form factor capable of wrist and fingertip measurements.**

## I. Introduction

Blood oxygen is critical for assessing respiratory function and tissue oxygenation [1]. However, blood oxygen measurements have been associated with differences in skin-tone [2]. Current pulse oximeters utilize reflected or transmissive photoplethysmography (PPG) to measure deoxygenated and oxygenated blood, as these biomarkers have different absorption characteristics [1]. However, pulse oximeter performance varies across skin tones, with occult hypoxemia occurring more frequently in individuals with darker skin tones than in those with lighter skin tones [2].

DermaGlow is a recently introduced multi-wavelength wearable sensor that noninvasively quantifies skin-tone using visible and near infrared light spectroscopy [3]. However, the device is limited by its 0.3 Hz sampling rate, which is required to preserve signal integrity, but is too slow to accurately capture $SpO_2$ measurements (typically 1-256Hz and 10-25Hz in clinical settings [4]). Thus, we present a device that integrates DermaGlow with PPG signals to provide complementary information to characterize skin-tone dependent variations in optical measurements.

## II. Device Hardware

We designed a device combining DermaGlow (skin-tone quantification system) with an $SpO_2$ sensor (Fig. 1). The skin-tone quantification system collects 16-bit precision spectral data at a sampling rate of 0.3Hz and includes three subsystems: The LEDs, the LED Driver, and an 11-channel photodetector array. The 11 LED wavelengths range from 415-940nm with ∼30nm spacing. The LED driver uses PWM dimming to control the brightness of the LEDs. Lastly, the photodetector array features eight visible photodetectors, one infrared photodetector, and a broadband clear channel for normalization and calibration. The $SpO_2$ sensor consists of the MAX30101 and MAX32664 from Analog Devices for raw PPG acquisition and $SpO_2$ estimation. The MAX30101 is the spectral $SpO_2$ sensor with a 10.24MHz sampling rate;

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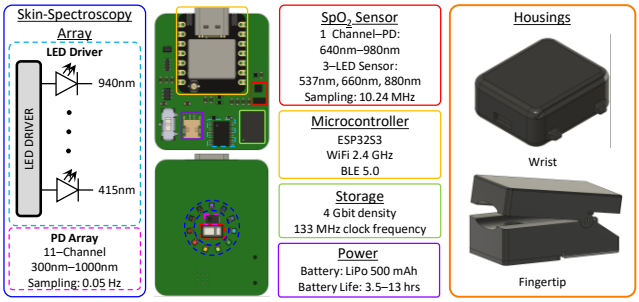



Fig. 1. Device Design

three LEDs with center wavelengths at 537, 660, and 880nm; and a single photodiode with a spectral range of 640-980nm.

## III. Results

We were able to design a single device that combines the hardware from DermaGlow and an off the shelf pulse oximeter for both wrist worn and fingertip use. However, initial hardware testing and debugging showed failed $I^2C$ communication with the $SpO_2$ sensor due to disconnected MFIO and Rest pins from the MAX32664 and the microcontroller. A new design iteration of the printed circuit board correctly connects the MFIO and Reset pins. Furthermore, the sensing suite is slightly large for fingertip data collection for individuals with smaller hand sizes. Investigation towards decreasing sensor suite size is addressed in Section IV.

## IV. Conclusion & Future Work

We were able to combine the hardware of an $SpO_2$ sensor with the hardware used in DermaGlow that fits in both fingertip and wrist worn form factors. While the device hardware requires minor adjustments, the firmware and algorithms required to continuously monitor $SpO_2$ while accounting for skin-tone attributes still need to be developed. Furthermore, investigating the LED/PD pairs most relevant for skin-tone quantification may reduce sensing suite size, while investigating confounding factors such as motion artifacts may reduce pulse oximeter error. Lastly, we will conduct a controlled desaturation study to evaluate device performance under hypoxic conditions.

## References

[1] A. Jubran, "Pulse oximetry," *Critical care*, 2015.
[2] M. W. Sjoding, R. P. Dickson, T. J. Iwashyna, S. E. Gay, and T. S. Valley, "Racial bias in pulse oximetry measurement," *New England Journal of Medicine*, 2020.
[3] T. Hamid, P. Flores, J. Byun, E. Courtney, K. Quinn, and A. Watson, "Dermaglow: Objective quantification of melanin, erythema and skin-tone using wearable optical spectroscopy," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 2025.
[4] T. Leppänen, S. Kainulainen, H. Korkalainen, S. Sillanmäki, A. Kulkas, J. Töyräs, and S. Nikkonen, "Pulse oximetry: the working principle, signal formation, and applications," in *Advances in the Diagnosis and Treatment of Sleep Apnea: Filling the Gap Between Physicians and Engineers*. Springer, 2022.